Arsh Singh

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Skills

- **Languages**: C++, Java, C, Python, JavaScript, SQL
- **Frameworks**: Streamlit, TensorFlow, Keras, Django, React, HTML, CSS
- **Tools/Platforms**: MySQL, GitHub, Jupyter Notebook, Google Colab, VS Code, Linux, Windows
- **Soft Skills**: Problem-Solving Skills, Project Management, Adaptability, Time Management

Projects

- **AI Based Job Recommendatio System** | Github — Jan 2026 - Mar 2026
  - Developed an AI-based job recommendation system that extracts and analyzes CV data (skills, education, experience) using NLP techniques and processes PDF resumes into structured information.
  - Implemented machine learning and semantic similarity models to match candidate profiles with job descriptions and rank them based on skill relevance, salary potential, and location proximity.
  - Designed features like resume scoring, top-N job recommendations, and efficient data handling using Python, Scikit-learn, and Sentence Transformers for a scalable real-world solution.

  Tools & Tech:Python, NumPy, Pandas, Scikit-learn, Sentence Transformers, PyPDF2, Streamlit, HTML/CSS, MySQL, Git, Jupyter Notebook
- **Anomaly Detection In Hyperspectral Image Processing** | Github — Apr 2025 - Jun 2025
  - Developed an end-to-end machine learning pipeline for anomaly detection in high-dimensional hyperspectral image data, addressing challenges such as large spectral bands, noise, and data redundancy for applications in agriculture, remote sensing, and environmental monitoring.
  - Carried out detailed preprocessing steps including data normalization, noise reduction, and PCA-based dimensionality reduction to enhance data quality, reduce computational complexity, and improve model performance.
  - Implemented and analyzed multiple anomaly detection methods such as One-Class SVM, Isolation Forest, Autoencoders, and Spectral Angle Mapper (SAM), and evaluated results using precision, recall, F1-score, and visual anomaly mapping to achieve accurate and efficient detection of anomalous regions.

  Tools & Tech:Python, NumPy, Pandas, Scikit-learn, TensorFlow/Keras, Matplotlib, Seaborn, OpenCV, Jupyter Notebook

Training

- **Summer Internship Training** | certificate — Aug 2025
  - Completed an intensive 7-week training in Advanced Python, Data Science, and Machine Learning, focusing on real-world problem solving and industry-oriented workflows.
  - Acquired hands-on experience in Python programming, data cleaning, preprocessing, feature engineering, model training, optimization, and performance evaluation using real datasets.
  - Implemented a wide range of machine learning techniques, including Linear & Logistic Regression, Decision Trees, Naïve Bayes, SVM, and Clustering, while extensively using NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn for data analysis and visualization.

  Tools & Tech:Python, NumPy, Pandas, Seaborn, and Scikit-learn for advanced data analysis, visualization,Matplotlib, and machine learning model development

Certificates

- Advanced Python for ML & AI - CSE Pathshala — Aug 2025
- Social Networks - NPTEL Online Certification — Apr 2025
- Computer Communications - Coursera — Nov 2024
- The Bits and Bytes of Computer Networking - Google — Sept 2024

Achievements

- **Code Crafter Smart India Hakathon 3rd rank** | certificate — Oct 2024

  Led team "Void Walkers" to secure 3rd place against 60 teams from different universities.

Education

- **Lovely Professional University** — Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 7.20*** — Aug 2023 - Present
- **Gajera International School** — Surat, Gujarat
  Intermediate; ***Percentage: 74%*** — Apr 2020 - Mar 2022
- **Vigyanand Kendriya Vidyalaya** — Siwan, Bihar
  Matriculation; ***Percentage: 93%*** — Apr 2019 - Mar 2020